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INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Autumn Semester Examination 2024-25

Date of Examination: 20th Nov 2024 Session: AN Duration: 3 Hrs Full Marks: 80

Subject No.: AI61004 Subject : Statistical Foundations of AI and ML Department: AI

Specific charts, graph paper, log book etc., required _____

Special Instructions (if any) : _____

ANSWER ANY 4 QUESTIONS. ALL PARTS OF SAME QUESTION MUST BE ANSWERED TOGETHER.

Use the Pseudorandom Number sequence (0-99) below for any sampling purposes.

62	4	99	60	12	91	21	79	33	71	94	47	10	48	29	72	68	41	73	62
18	88	91	27	42	21	55	51	8	87	19	86	79	14	13	95	7	76	53	41
14	54	32	67	21	87	73	12	38	98										

Q1. A) Consider a short sequence of observations [3.6 7.5 2.4]. Based on AIC, which of the following models would you choose to fit these observations?

- i) M1: a GMM with 2 states with $\pi=[0.6, 0.4]$, $\mu=[3, 9]$, $\sigma^2=[1, 4]$
- ii) M2: a HMM with 2 states with $\pi=[0.5, 0.5]$, $A=[[0.6, 0.4],[0.4, 0.6]]$, $\mu=[3, 7.5]$, $\sigma^2=[1, 1]$

According to the chosen model, calculate the probability that the second observation (7.5) was generated by the second state, i.e. $Z(2)=2$. **[4+4=8 marks]**

B) Consider you have N sequences of length T each, in the form $\{\{x_{11}, \dots, x_{1T}\}, \{x_{21}, \dots, x_{2T}\}, \dots, \{x_{N1}, \dots, x_{NT}\}\}$, where x_{nt} are integers in $\{1, 2, \dots, L\}$. You want to fit a K -state Hidden Markov Model on this dataset. Considering that the emission distribution is Categorical,

- i) Write down the log-likelihood function in terms of all model parameters **[4 marks]**
- ii) Derive a tight lower bound for this log-likelihood with explanations **[4 marks]**
- iii) Write down the update equations for all model parameters for the M-step. **[4 marks]**

Q2. Consider the following latent variable model: $Z_i \sim \text{Cat}(\pi)$, $Y_i \sim U(a_k, b_k)$, $X_i \sim N(Y_i, v_k)$ where $k = Z_i$. Clearly, the model parameters are $[\pi, \{a, b, v\}_k]$. The observations are $[x_1, x_2, \dots, x_N]$ (all real numbers), while Y, Z are latent. Y are real-valued while $Z \in \{1, 2, \dots, K\}$.

- i) Write the expression for the model log-likelihood **[4 marks]**
- ii) Discuss how the parameters may be estimated using E-M algorithm. Explain what will be done in the E-step and M-step. **[8 marks]**

- iii) Derive $p(Y_i | X_i, Z_i)$ and $p(Z_i | X_i, Y_i)$ in terms of the parameters. [4 marks]
- iv) Suppose Y, Z are known along with X . Derive ML estimates of all parameters. [4 marks]

Q3. A) You have a dataset X of 10 binary observations. Your Null Hypothesis is that the Bernoulli parameter is 0.5, while the alternate hypothesis is that this parameter is above 0.5. Your test-statistic $T(X)$ is the fraction of '1's, and your rejection region is defined by $(T(X) - 0.5) > \epsilon$. What will be possible values of ϵ , if you are to reject the null hypothesis at 5% level of significance? [4 marks]

B) Draw 5 samples $(x_1, x_2, x_3, x_4, x_5)$ from $N(5, 10)$ and another 5 more samples $(y_1, y_2, y_3, y_4, y_5)$ where $y_i \sim N(x_i, 10)$. Carry out a permutation test to check $(x_1, \dots, x_5)$ and $(y_1, \dots, y_5)$ belong to the same distribution. [6+6=12 marks]

C) You have N real-valued observations. Explain how you can test a null hypothesis that the observations were obtained from a Gaussian Mixture Model. [4 marks]

Q4. A) Consider a PDF $f(x) = 0.3*f_1(x) + 0.4*f_2(x) + 0.3*f_3(x)$ where $f_1(x) = x^5*(1-x)^3/B(6, 4)$, $f_2(x) = 1$ and $f_3(x) = x^3*(1-x)^5/B(4, 6)$ where $x \in (0,1)$. [$B(6,4) = B(4,6) = 0.002$]

- i) Use rejection sampling to draw a sample from $f(x)$. Use suitable envelope and the pseudorandom number sequence on top. [6 marks]
- ii) Demonstrate Metropolis-Hastings Algorithm to draw 2 samples from $f(x)$ with suitable proposal distribution and the pseudorandom number sequence on top. [6 marks]
- iii) Explain in the above case, what happens if the proposal distribution had very small variance. [4 marks]

B) Explain the burn-in phase and its significance in Metropolis-Hastings algorithm. [4 marks]

Q5. A) Prove that PDF for chi-square distribution with degree k is Gamma $(k/2, 1/2)$. [8 marks]

B) Derive the posterior distribution for the "precision" parameter of a Gaussian distribution with known mean, with Gamma prior. [8 marks]

C) Explain why E-M algorithm is guaranteed to converge to at least a locally optimal point in the parameter space. [4 marks]